

ARITI CHAWLA

AI/ML Engineer | Data Scientist | Full-Stack Developer

Delhi, India | aritichawla2110@gmail.com | +91-9773804367 | linkedin.com/in/ariti-chawla |
github.com/ariti2110 | portfolio-theta-beryl-86.vercel.app

CORE COMPETENCIES

Python • Machine Learning (XGBoost, scikit-learn, TensorFlow, Keras) • LangChain • RAG • Retrieval-Augmented Generation • NLP • Large Language Models (LLM) • Prompt Engineering • Generative AI • Data Analysis (pandas, NumPy) • Statistical Analysis • Feature Engineering • Model Explainability (SHAP, LIME) • Model Ensemble • Hyperparameter Tuning • Cross-validation • A/B Testing • Experimentation • SQL • ETL Pipelines • Data Preprocessing • EDA (Exploratory Data Analysis) • PostgreSQL • MongoDB • Node.js • Express.js • React.js • FastAPI • REST APIs • JWT Authentication • API Development • Microservices • Power BI • Tableau • Dashboard Development • Business Intelligence • Git • GitHub • Agile/Scrum • Vector Databases • Time Series Analysis • Predictive Modeling

PRODUCTION AI SYSTEMS & KEY PROJECTS

SHL Assessor Bot — Full-Stack Conversational AI | Production (Replit Autoscale)

End-to-end LLM orchestration: LangChain + Claude API + Chainlit + Replit Autoscale. Advanced prompt engineering for multi-turn conversations, context window management (token optimization), conversation state persistence, hallucination prevention. Production deployment handling live users with error recovery & graceful degradation. BM25-based retrieval pipeline, full README documentation. Demonstrates complete AI system engineering: architecture design, prompt optimization, infrastructure scaling, real-time performance monitoring.

FarmAI — National Hackathon Winner (Defeated 500+ Teams)

AI-powered agricultural intelligence: LangChain + Google Gemini Pro + ML ensemble for intelligent crop yield prediction. 88% prediction accuracy across 12+ crop varieties. RAG-like architecture: ingest agricultural datasets → retrieve relevant ML models → synthesize multi-crop recommendations. Natural language interface enabling farmers to query in plain language (NLP application, reducing technical friction). Real user validation & deployment. Won Geek Room Case Clinic, ₹10,000 prize. Demonstrates: Generative AI, NLP, LLM integration, AI product thinking, rapid prototyping & validation.

PROFESSIONAL EXPERIENCE

DRDO Scientific Analysis Group, Ministry of Defence — ML Engineer (Dec 2025–May 2026)

Engineered end-to-end ML system for cryptanalysis: processed 1M+ RC4 keystream samples | 95% vulnerability detection accuracy (security-critical threshold) | XGBoost + Artificial Neural Network ensemble with SHAP/LIME explainability for cryptanalytic insights | 3× turnaround improvement (2 weeks → 5 days). Full ETL pipeline: data cleaning, byte-level feature engineering, statistical bias analysis. Production deployment for Ministry of Defence research team. Keywords: Machine learning pipeline, Model ensemble, Explainable AI, Statistical analysis, Performance optimization, Cryptanalysis, Government research.

Infosys Springboard — Crypto Portfolio Manager, Full-Stack AI (Feb–Apr 2026)

Full-stack system: Python backend (FastAPI) + Node.js/Express + React frontend + PostgreSQL. JWT authentication, real-time KPI dashboards (50+ digital assets), AI-driven investment recommendation engine. 18% outperformance vs buy-and-hold baseline (validated through backtesting). 8 concurrent users. 30% improvement in requirements-to-delivery cycle. Microservices architecture, API development, Database optimization, Recommendation systems, Time series analysis.

Reve Cult — Marketing Analytics Engineer (Sep–Dec 2025) | Intern of the Month

Designed Power BI dashboards tracking campaign KPIs (CTR, conversion rate, reach, ROI) across 5 product lines. Optimized 15+ data-driven campaigns through SQL analytics & A/B testing interpretation: 30%+ engagement improvement, 100K+ impressions, 25% ROI increase. Partnered with marketing & product teams on KPI-driven strategy. Dashboard development, Business Intelligence, Data analysis, SQL optimization.

AWARDS & CREDENTIALS

FarmAI Hackathon Winner (Geek Room Case Clinic, ₹10,000) • Google Prompt Creator Award (May 2026, Top Award) • Google Career Glow-Up Top 5 Finalist • McKinsey Forward Program (Leadership Development) • Coding Ninjas DSA Certificate of Excellence (Top Performer, Sep 2023–Mar 2024) • Coding Ninjas Introduction to Python Certificate of Excellence • Samsung Innovation Campus Big Data (Nov 2025–Mar 2026) • UiPath Automation Developer SAP Specialist • Tech Fest PRADARSHAN 4.0 – 2nd Place Mock Placement • LeetCode Data Navigator Badge • Reve Cult Intern of the Month Award (Dec 2025) • DRDO Ministry of Defence Completion Certificate (Trainee, SAG) • Infosys Springboard Completion Certificate • be10x AI Tools

& ChatGPT Workshop (Jan 2026) • British Airways Data Science Simulation (Forage, Feb 2025) • Tata Data Visualization Simulation (Forage, Jan 2025) • Modern AI Tools Certificate (Saras AI Institute, Aug 2025) • HackerRank Certificates (Python, Java, JavaScript, Problem Solving) • Skillsoft Leadership Development Program (MIT Sloan) • Certificate of Merit: Aspiring Academic Association Award

EDUCATION

B.Tech Computer Science Engineering — SGT University, Gurugram | Graduating 2026